物态

构成物质的分子的聚合状态称为物质的聚集态，简称物态。

气态

凝聚态：液态、固态

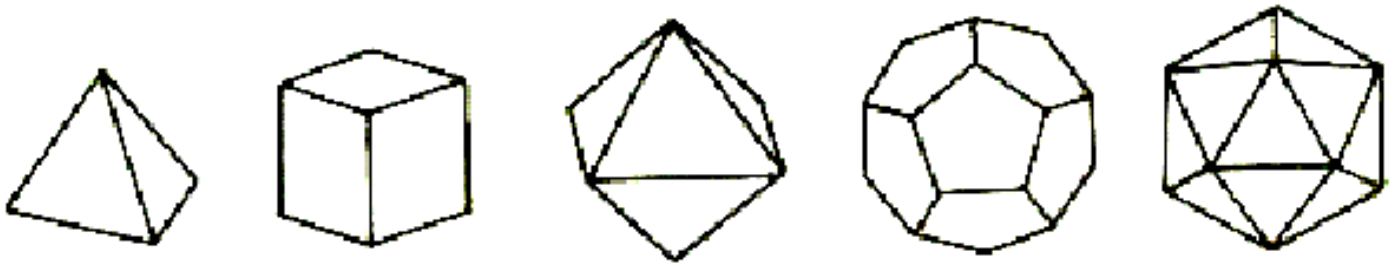
（外加：等离子态与超密态）

固体：晶体和非晶体

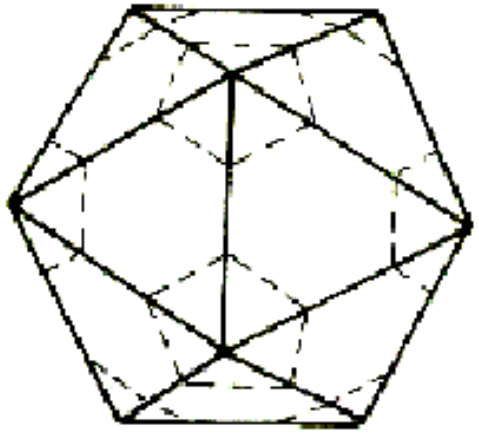
晶体：在整块晶体中沿各个方向晶体结构周期性　地、完整地重复。是长程有序结构。

单晶、多晶体

只要由同种材料制成，它在给定压强下的熔点、溶解热是确定。这是鉴别晶体、非晶体的最简单的方法。

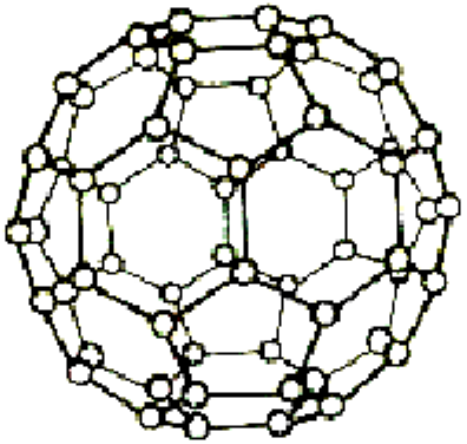


柏拉图体

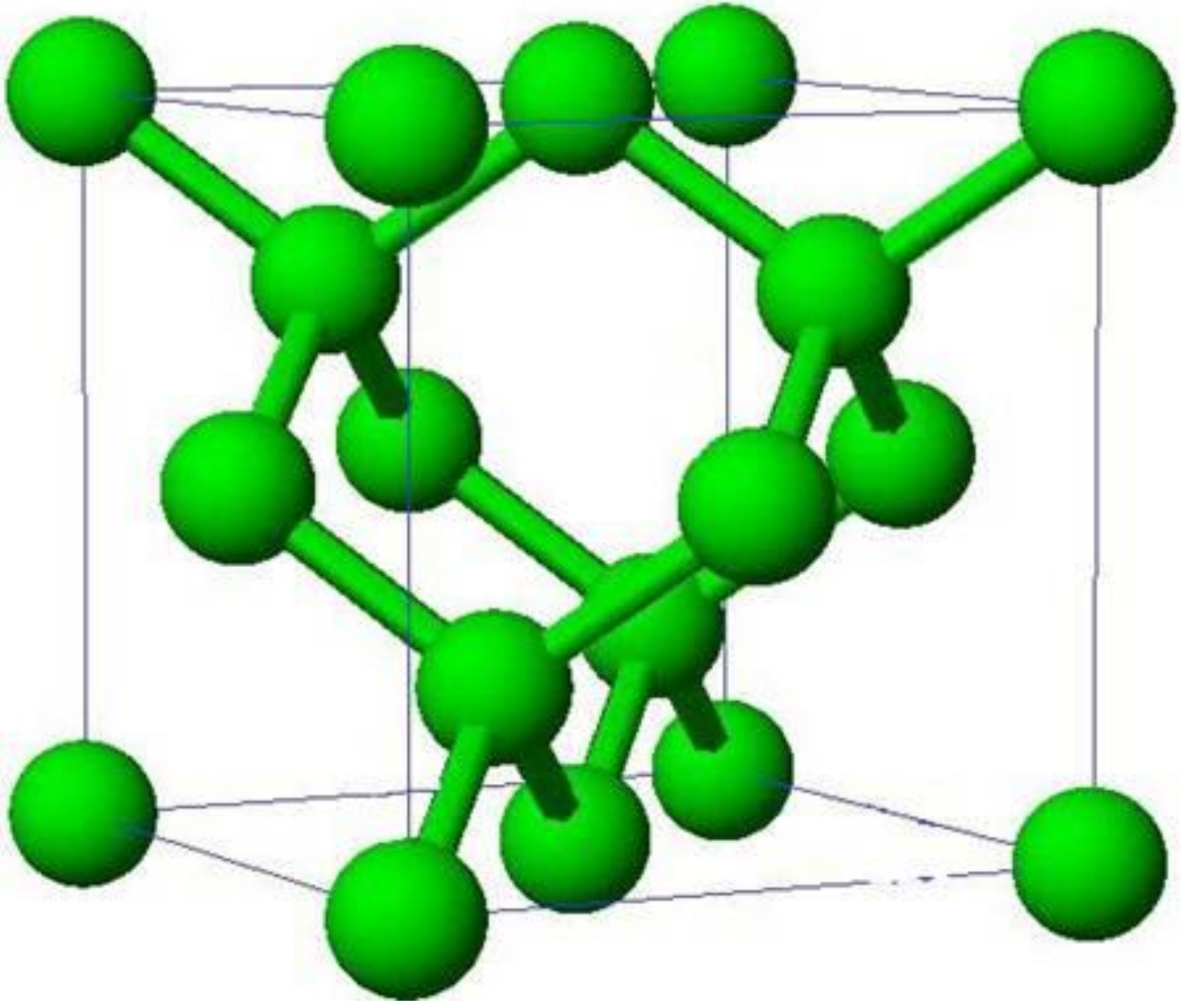


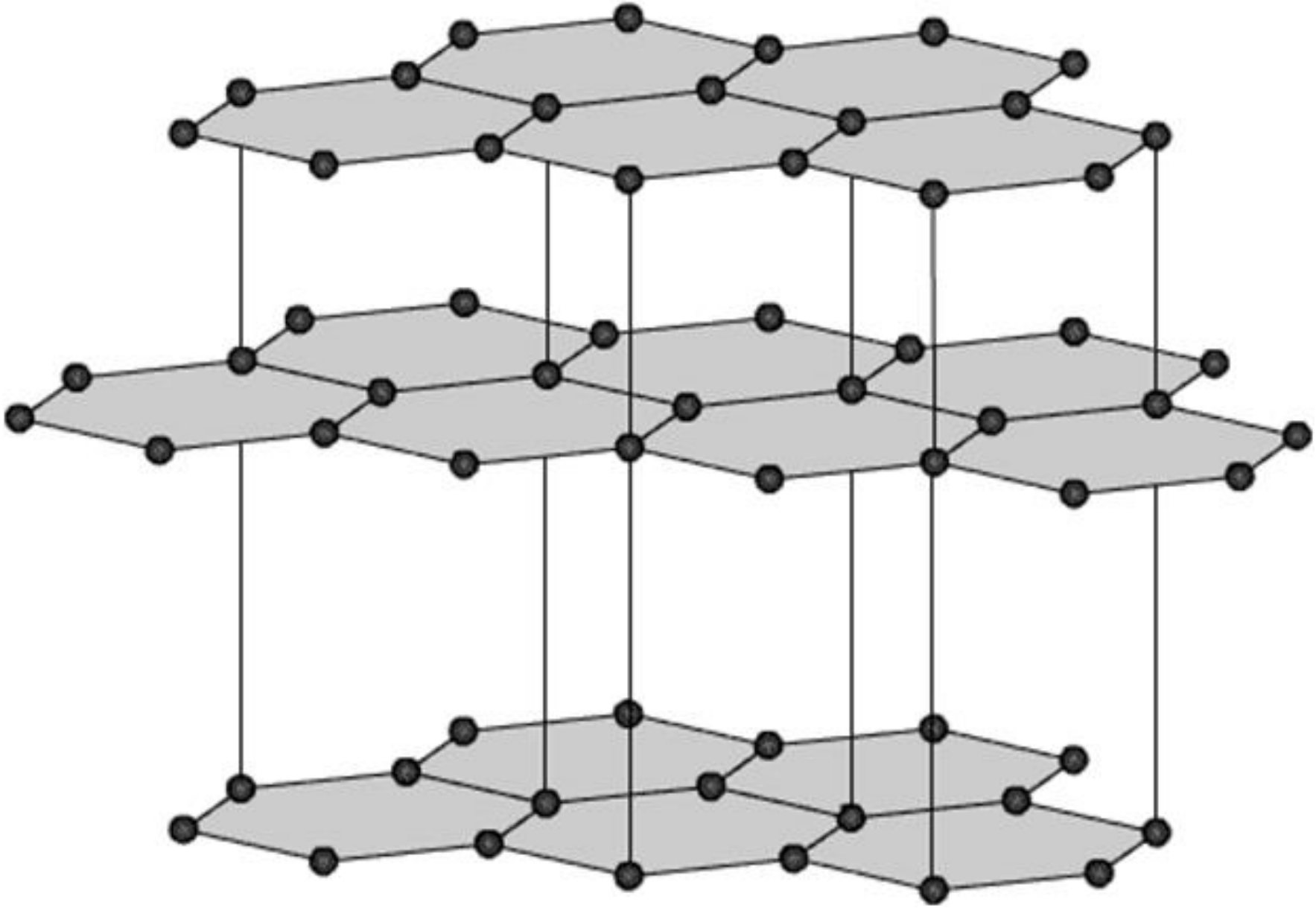
a. 截角二十面体

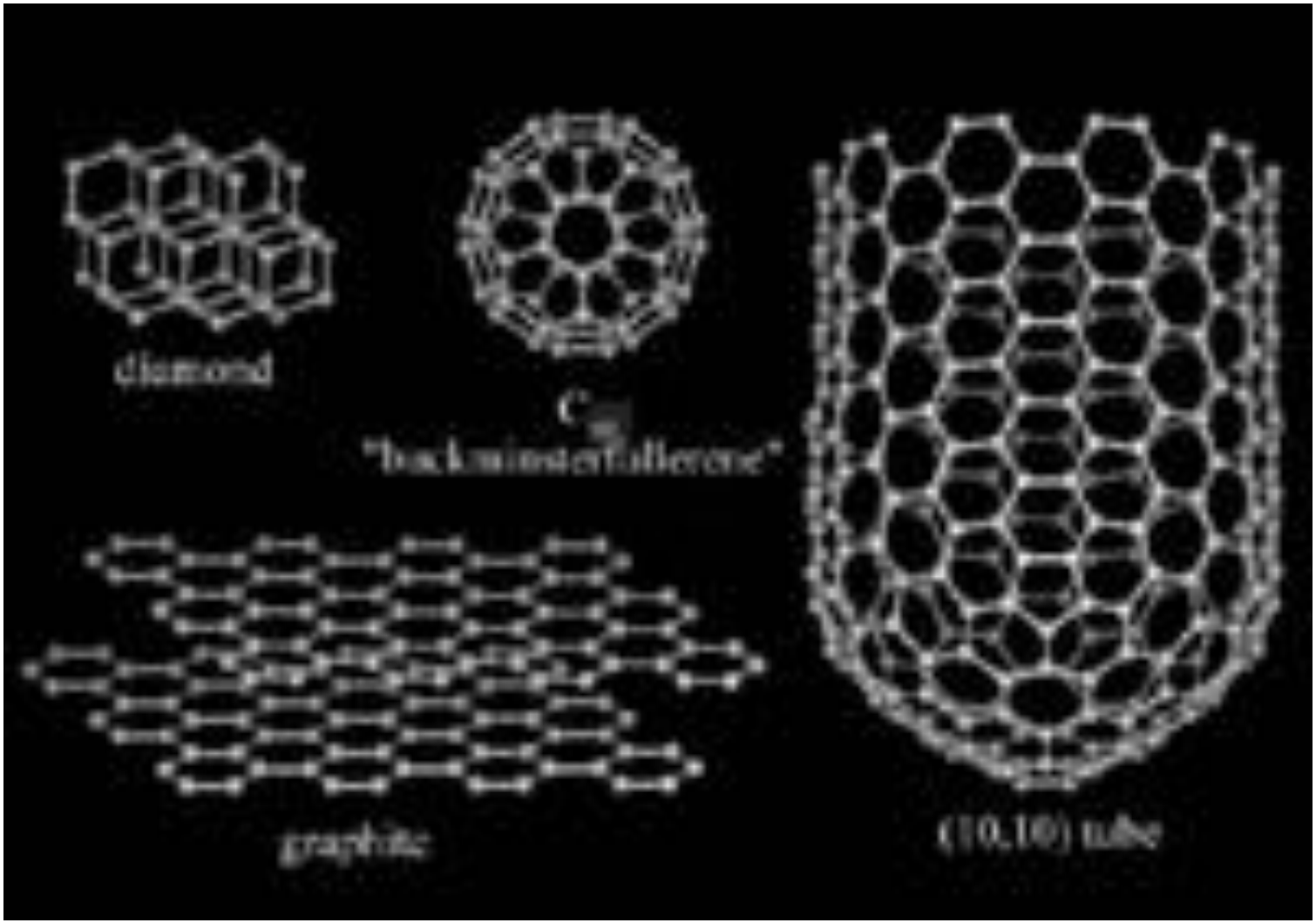
b. 足球



c. C_{60}分子






diamond
C_{60}
"buckminsterfullerene"
graphite
(10,10) tube

超密态物质

白矮星

A white dwarf star is at the center of the gas cloud.

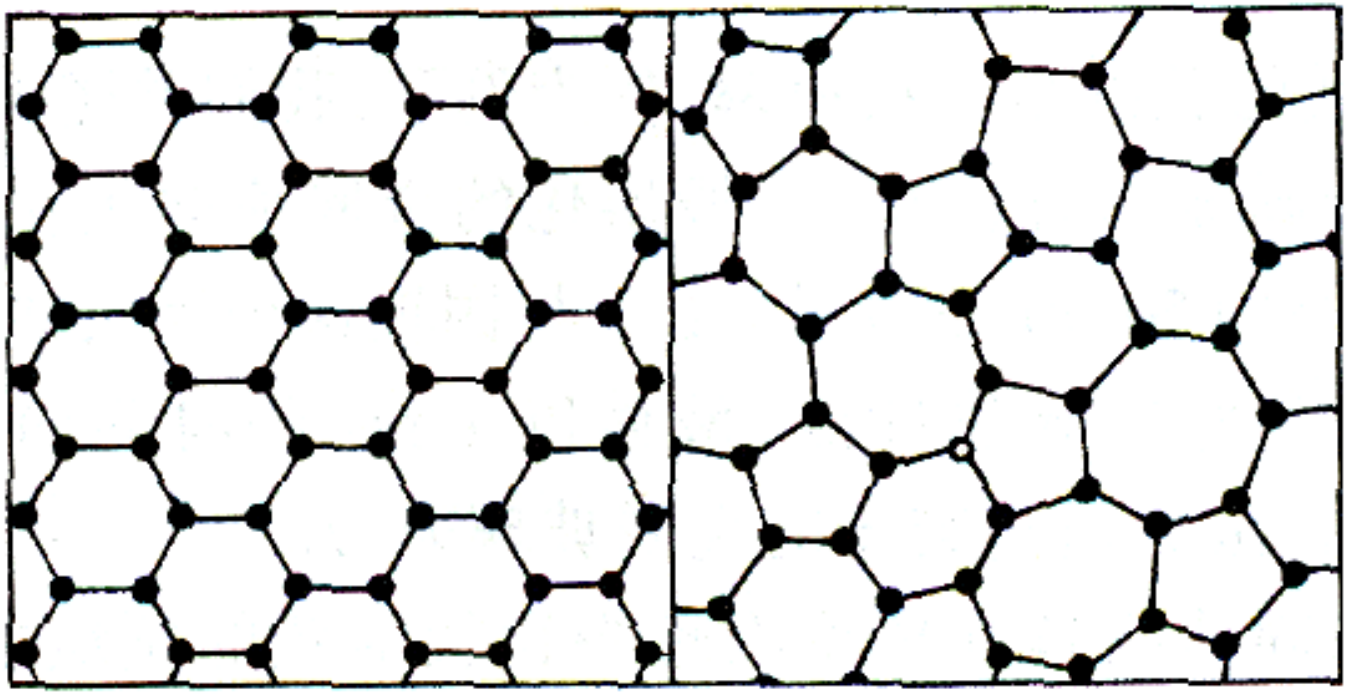
At the center of the crab nebula, shown here, is a neutron star, identifiable through its properties as a pulsar.

中子星

黑洞

液体：可以流动，其他很多特性与非晶体相像

如玻璃，短程有序，长程无序



a. 晶 体　　b. 玻璃体

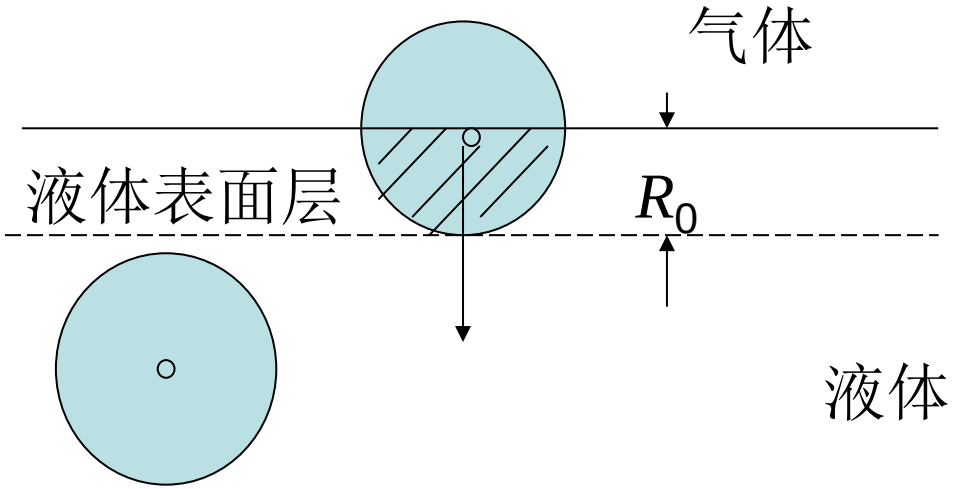
非晶态的短程有序和长程无序

*液体的表面现象

一、表面张力与表面能

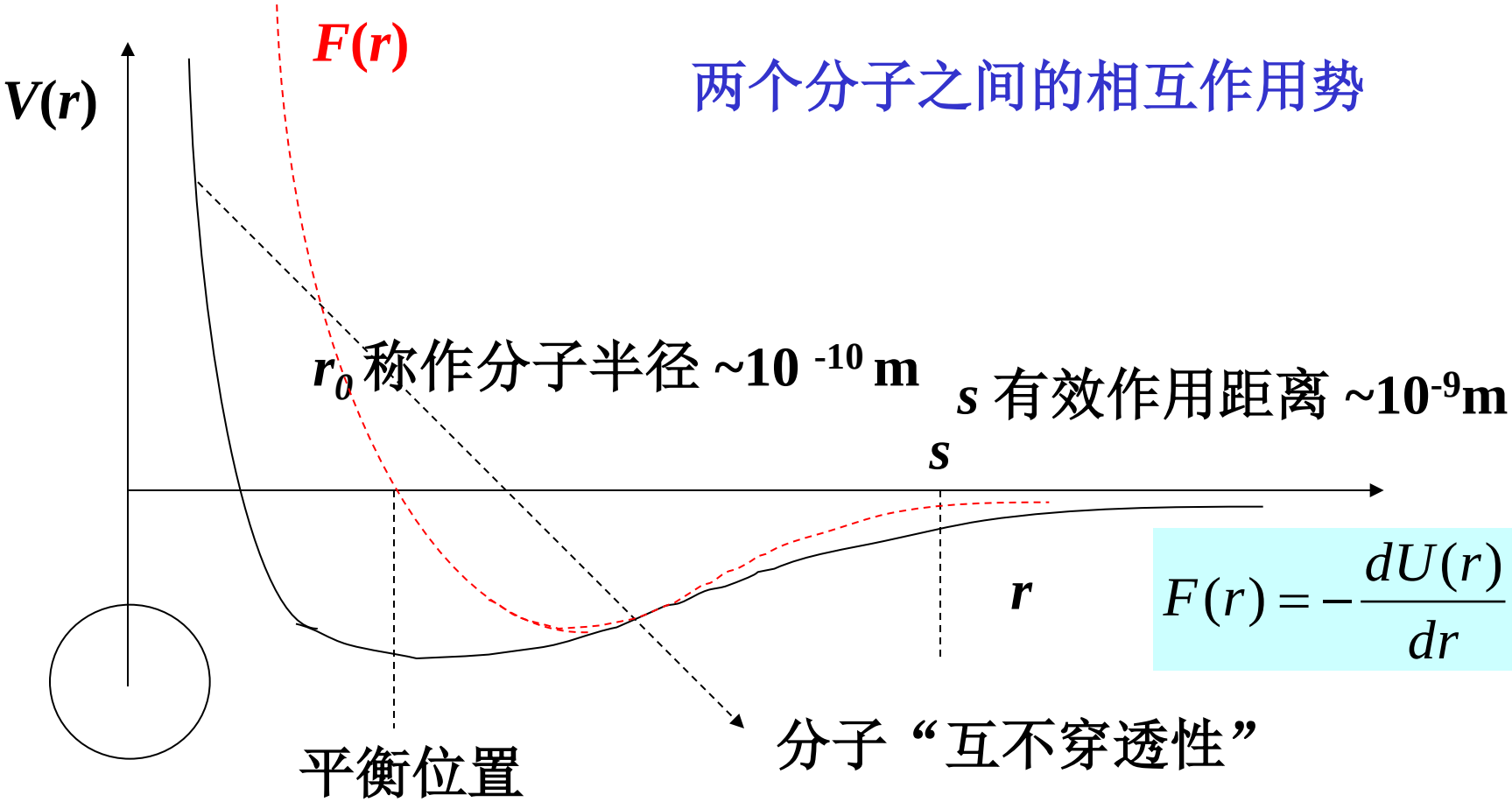
1、表面张力

表面张力是作用于液体表面的使液体具有收缩倾向的一种力。



液体表面更松散

两个分子之间的相互作用势

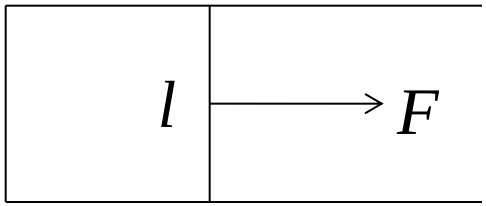


$$F(r) = -\frac{dU(r)}{dr}$$

热胀冷缩 — 平衡位置左右不对称

2、表面张力和表面张力系数

$$F = \sigma l$$



$$F = 2\sigma l$$

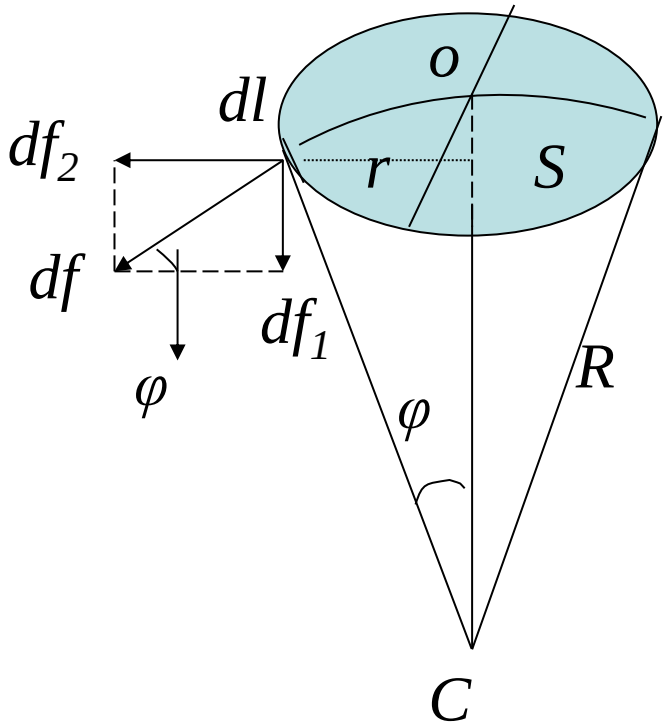
3、表面张力做功

$$dW = -\sigma dA$$

4、表面张力系数与温度的关系

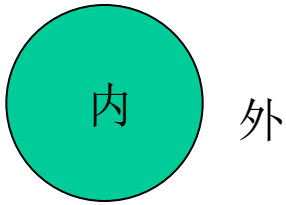
实验发现：液体的表面张力系数 σ 与液体的表面积的大小无关，而仅是温度的函数，表面张力随着温度的升高而降低。

二、弯曲液面附加压强

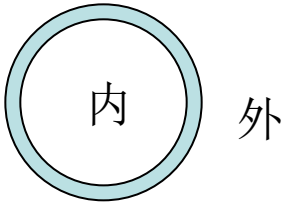


$$f_1 = \sigma 2\pi r \sin\varphi$$

$$P = \frac{f_1}{\pi r^2} = \sigma \frac{2\pi r}{\pi r^2}\frac{r}{R}$$



$$P_{内} = \frac{2\sigma}{R}$$



球形液膜 $P = \dfrac{4\sigma}{R}$

任意液面附加压强：$p=\sigma(\frac{1}{R_1}+\frac{1}{R_2})$

这是拉普拉斯公式

弹性曲面的附加压强：

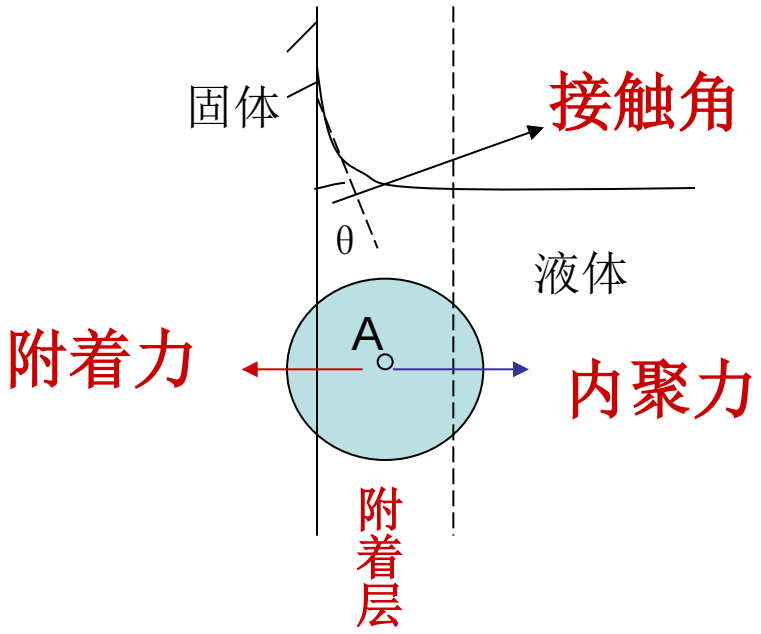
弹性管腔

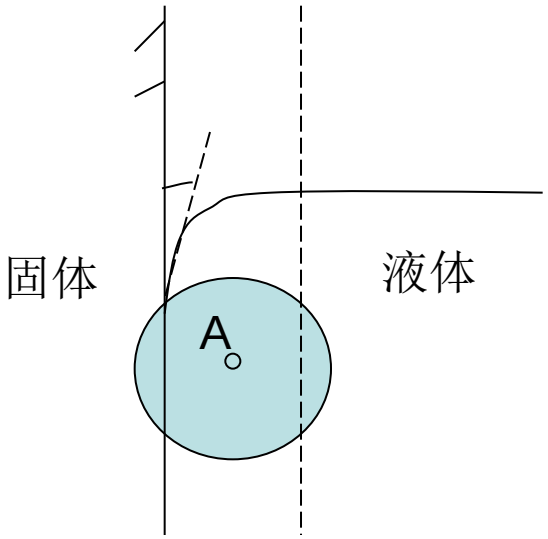
$$p=\frac{T}{R},\quad T\text{单位长度的膜张力}$$

三、润湿与不润湿　毛细现象

（一）润湿与不润湿

例如：水 +　玻璃　　水 + 涂有油脂的玻璃





（二）毛细现象

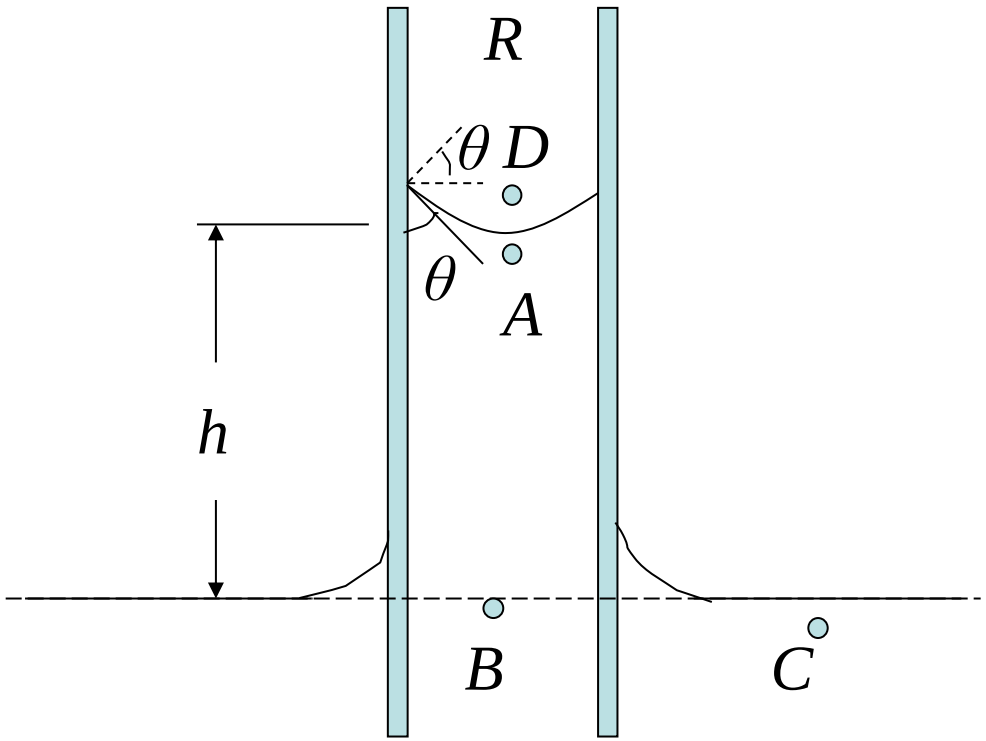
润湿

$$p_D = \frac{2\sigma}{R} + p_A = p_C = p_B$$

$$p_B = p_A + \rho g h$$

$$\rho g h = \frac{2\sigma}{R}$$

$$r = R\cos\theta$$



$$h = \frac{2\sigma\cos\theta}{\rho g r}$$

不润湿

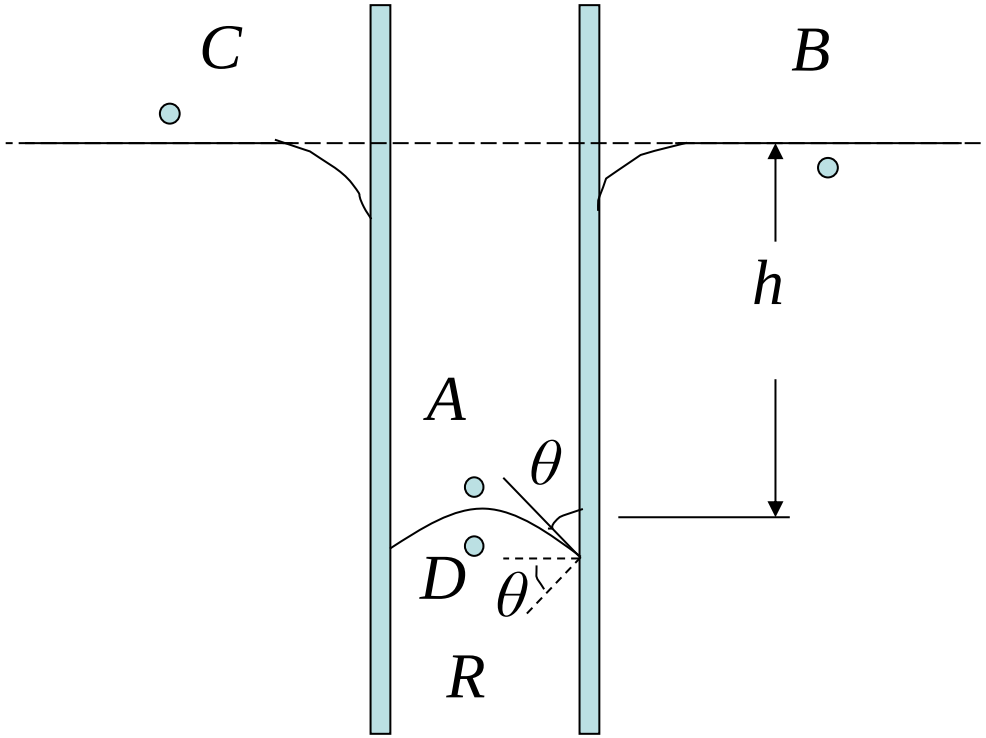
$$p_{\mathrm{D}} = \frac{2\sigma}{R} + p_{\mathrm{A}}$$

$$p_D = p_B + \rho g h$$

$$p_A = p_C = p_B$$

$$\rho g h = \frac{2\sigma}{R}$$

$$r = R\cos\theta$$



$$h = \frac{2\sigma\cos\theta}{\rho g r}$$

*相图

一、固、液、气相变

固---液相变：融解或凝固

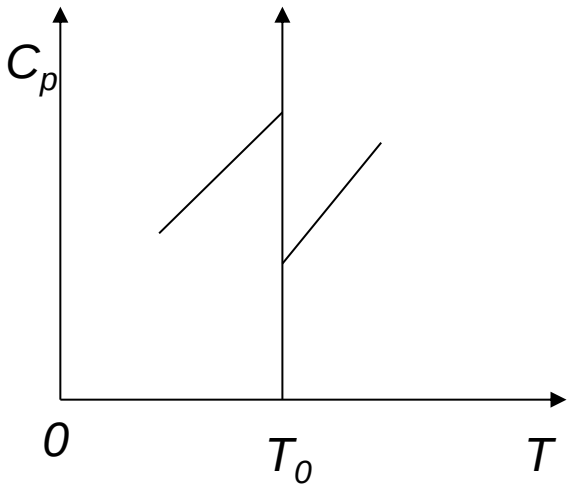
固---气相变：升华或凝华

液---气相变：气化或液化

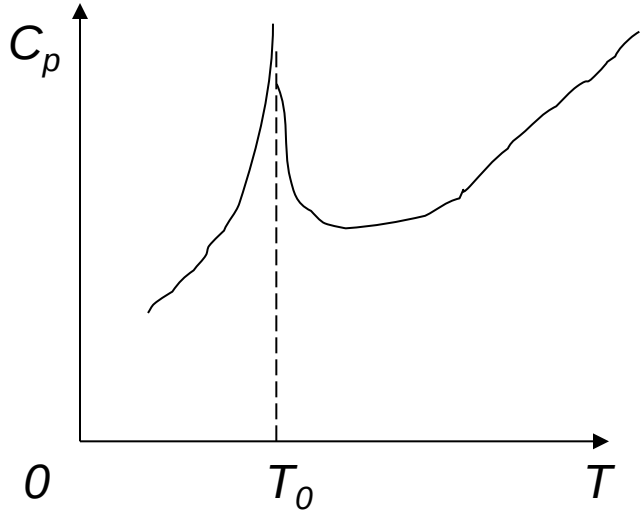
都伴随有潜热的吸放或体积的突变。

一级相变

一级相变和连续相变

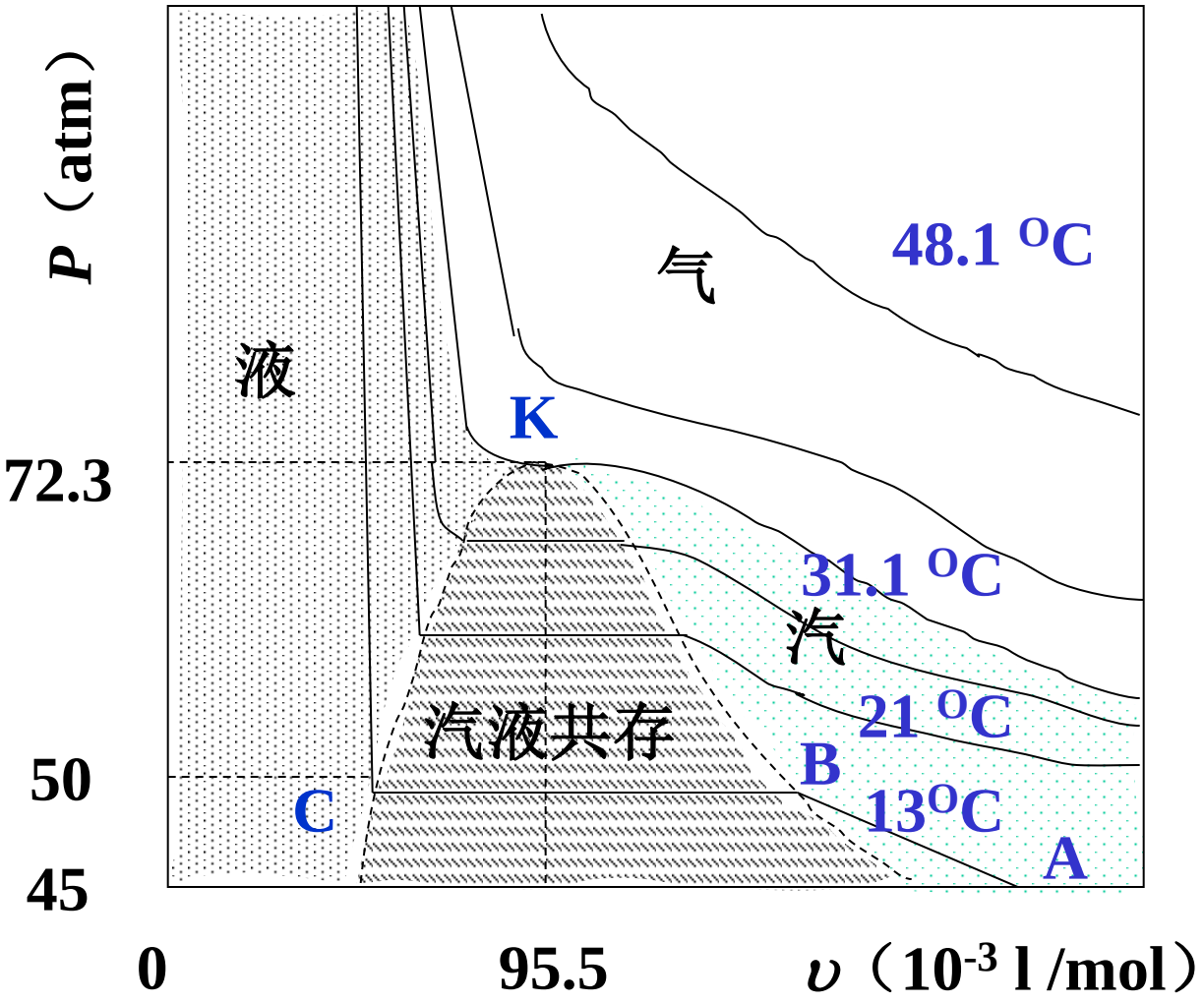


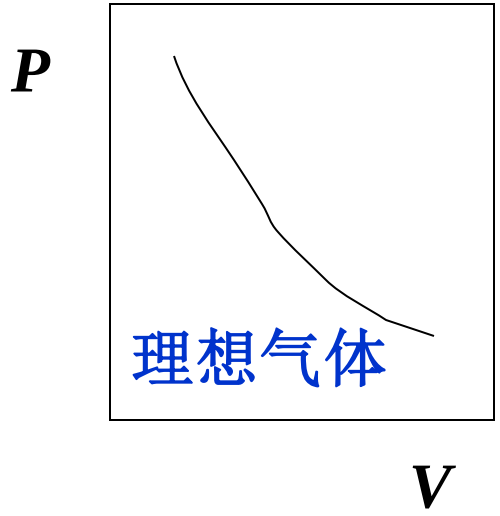
若对没有足够汽化核的液体加热，会产生过热液体；若对没有足够凝结核的蒸汽冷却，可形成过冷蒸汽，我们把摩尔熵和摩尔体积均发生突变的，可出现过冷、过热现象的相变称为一级相变。



发生相变时，相变前后两种相的摩尔熵和摩尔体积均不发生突变，这种相变称为连续相变，如：λ相变。

液汽相变　　真实气体等温线

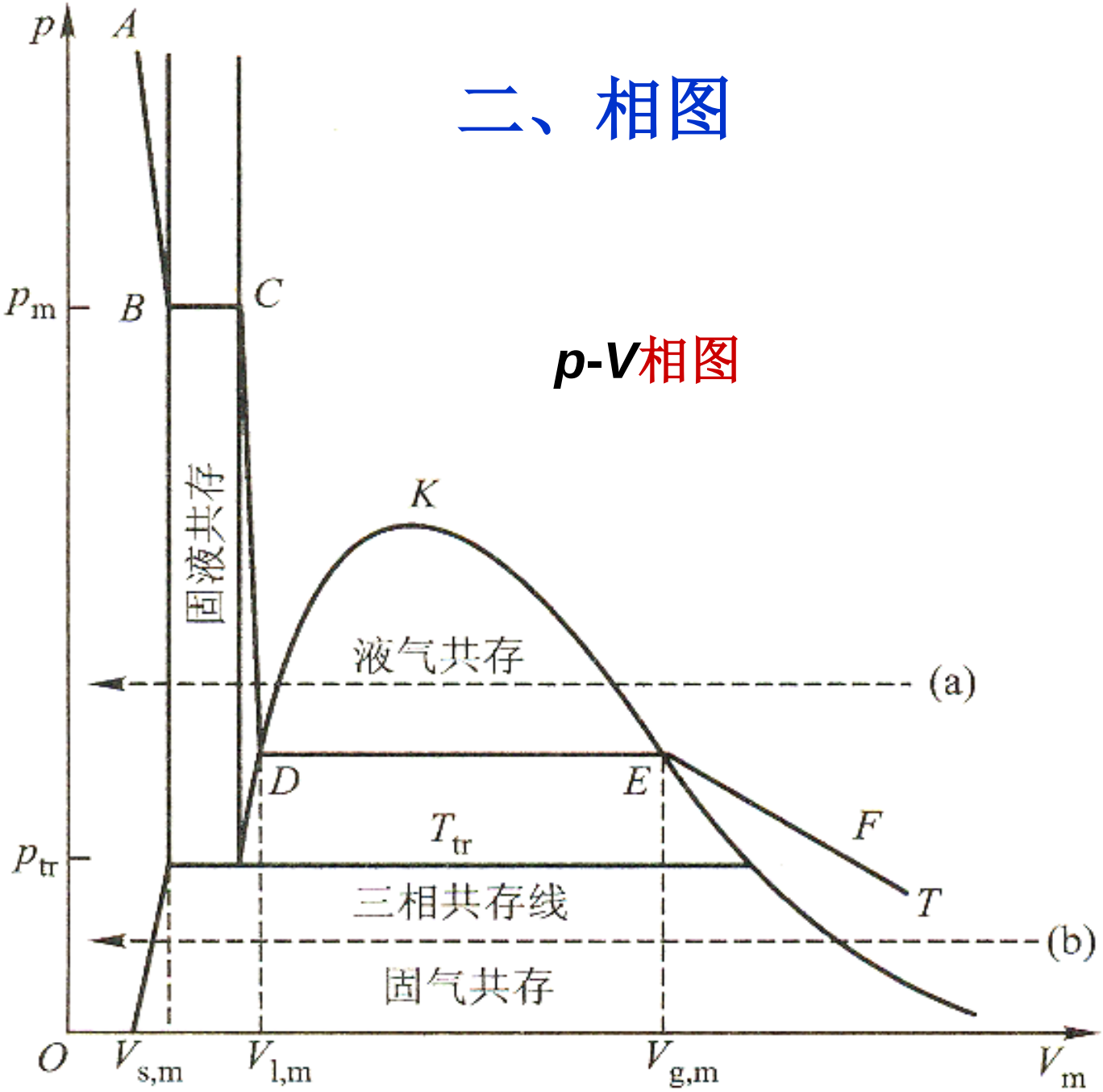




CO_2等温线

二、相图

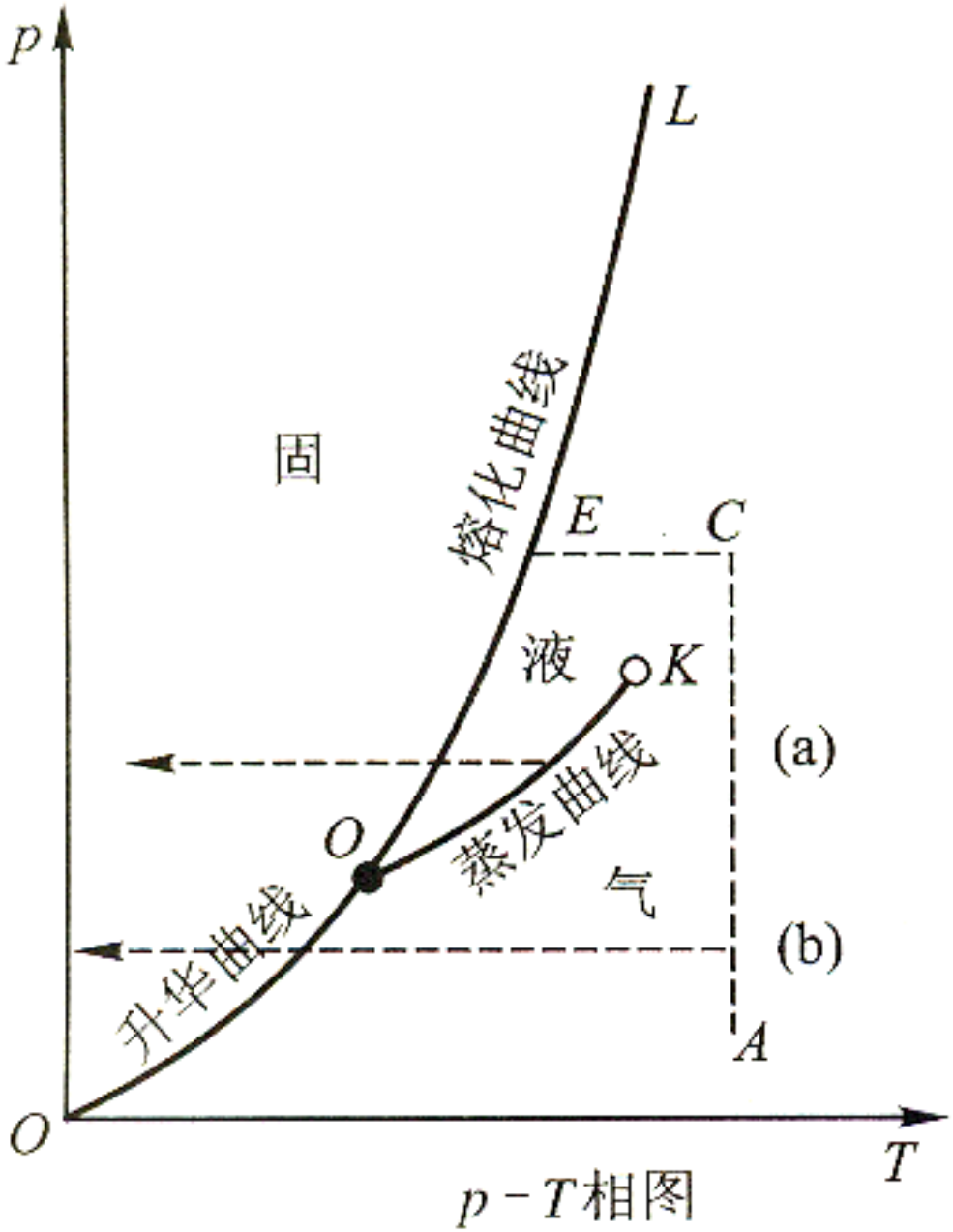
***p*-*V*相图**



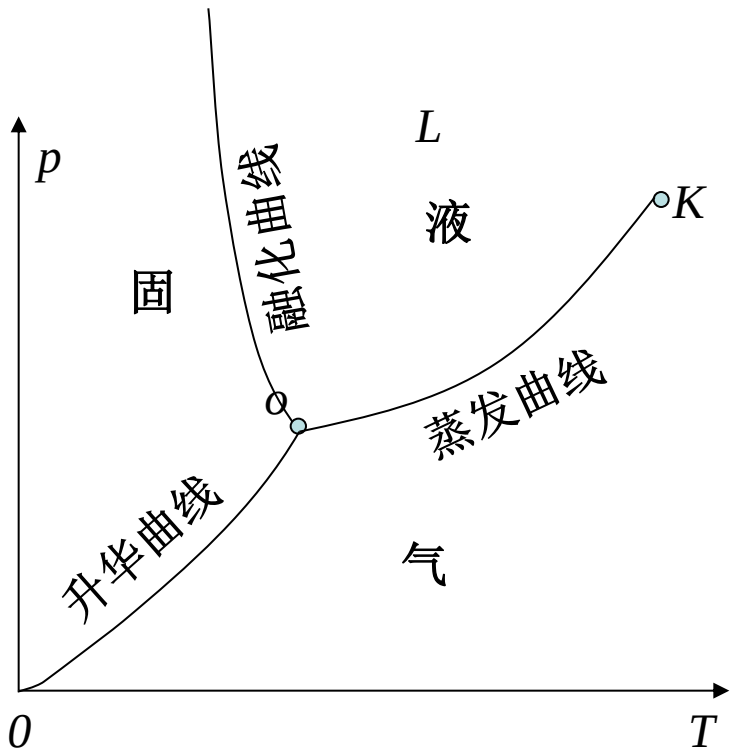
常见有***p-T***相图

三相点O

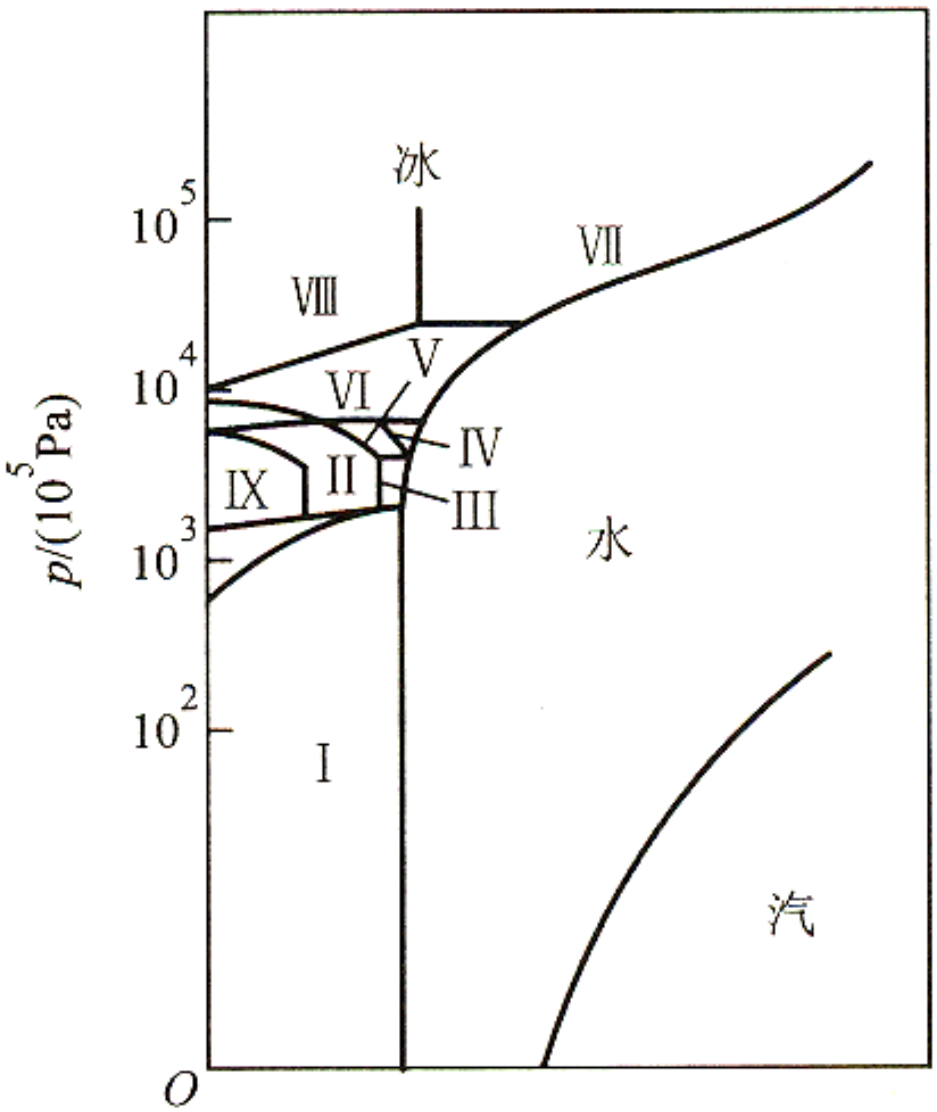
K点以上的区域
称为超临界状态



p－T相图



水的反常固液线



冰有9个相的相图

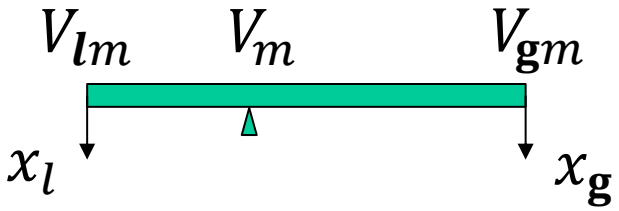
三、杠杆原理

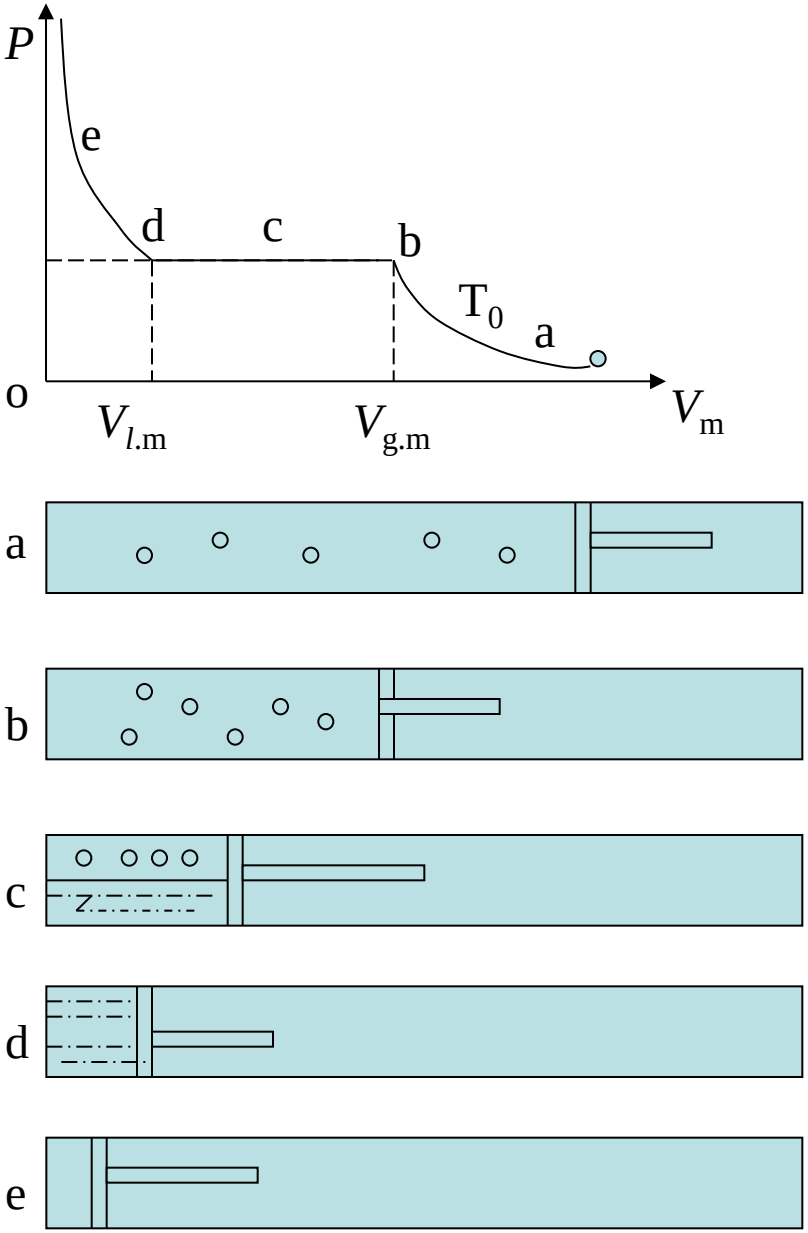
$$V_m = x_l V_{lm} + x_g V_{gm}$$

$$x_l + x_g = 1$$

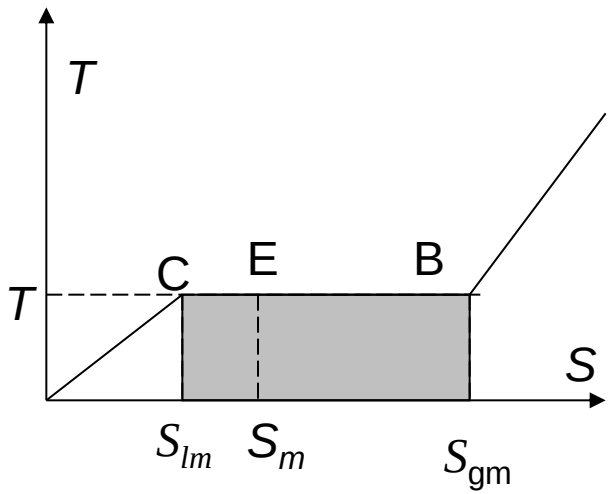
$$x_g = \frac{V_m - V_{lm}}{V_{gm} - V_{lm}}$$

$$x_l = \frac{V_{gm} - V_m}{V_{gm} - V_{lm}}$$





真实气体的*T-S*图与*p-V*图的比较



杠杆原理

$$x_g = \frac{S_m - S_{lm}}{S_{gm} - S_{lm}}$$

$$x_l = \frac{S_{gm} - S_m}{S_{gm} - S_{lm}}$$

$l_{v,m}$ 为摩尔汽化热

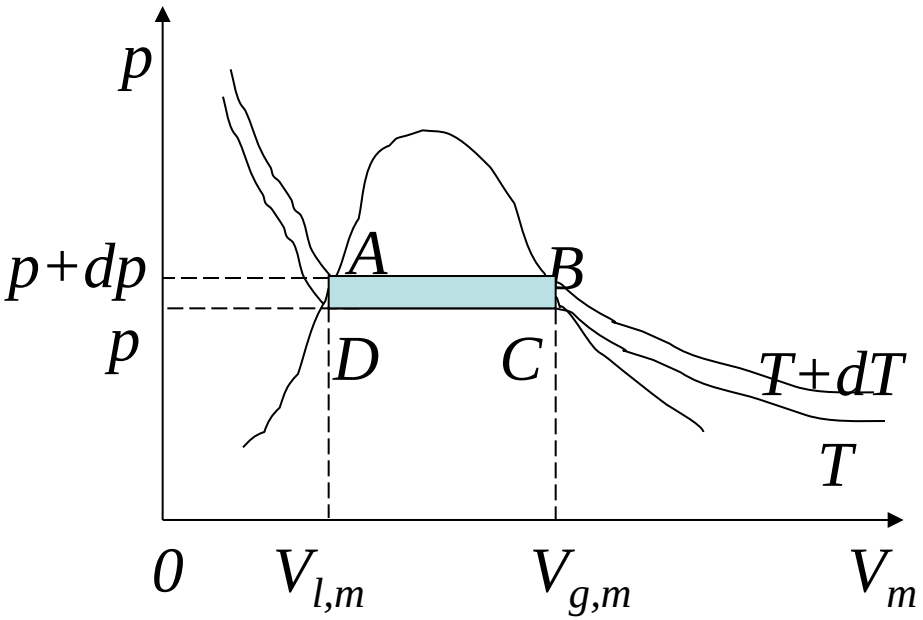
$$T(S_{g,m} - S_{l,m}) = l_{v,m}$$

汽化热

$$l_{v,m} = H_{g,m} - H_{l,m} = U_{g,m} - U_{l,m} + p_0(V_{g,m} - V_{l,m})$$

四、克拉珀龙方程

ABCDA组成可逆卡诺循环



$$\eta = \frac{dT}{T} = \frac{dp \cdot \nu (V_{g,m} - V_{l,m})}{\nu l_{\nu,m}}$$

$$\frac{dp}{dT} = \frac{l_{\nu,m}}{T(V_{g,m} - V_{l,m})}$$

克拉珀龙---克劳修斯方程

气---液两相 ⟶ 其它两相

$$\frac{dp}{dT}=\frac{l_{12,m}}{T(V_{1,m}-V_{2,m})}=\frac{l_{12}}{T(V_1-V_2)}$$

$$\because l_{12,m}=T(S_{1,m}-S_{2,m})$$

$$\therefore \frac{dp}{dT}=\frac{S_{1,m}-S_{2,m}}{V_{1,m}-V_{2,m}}$$

相图上相平衡曲线的斜率等于一级相变中摩尔熵和摩尔体积突变变量的比率。

蒸汽压方程：描述液---气及固----气的饱和蒸汽压随着相变温度变化的方程。

近似处理

- 温度变化不大时，汽化热或升华热不随温度变化
- 液相或固相的体积比气相少得多可忽略
- 饱和蒸汽压不大时，蒸汽可看作理想气体

$$\frac{dp}{dT}=\frac{l_{v,m}}{T(V_{g,m}-V_{l,m})}\approx\frac{l_{v,m}}{T(RT/p)}$$

$$\frac{dp}{p} = \frac{l_{v,m}}{R} \cdot \frac{dT}{T^2} \qquad p = C\exp(-\frac{l_{v,m}}{RT})$$

饱和蒸汽压随温度的增加而迅速增加

例 青藏高原上80度水开，问压强？

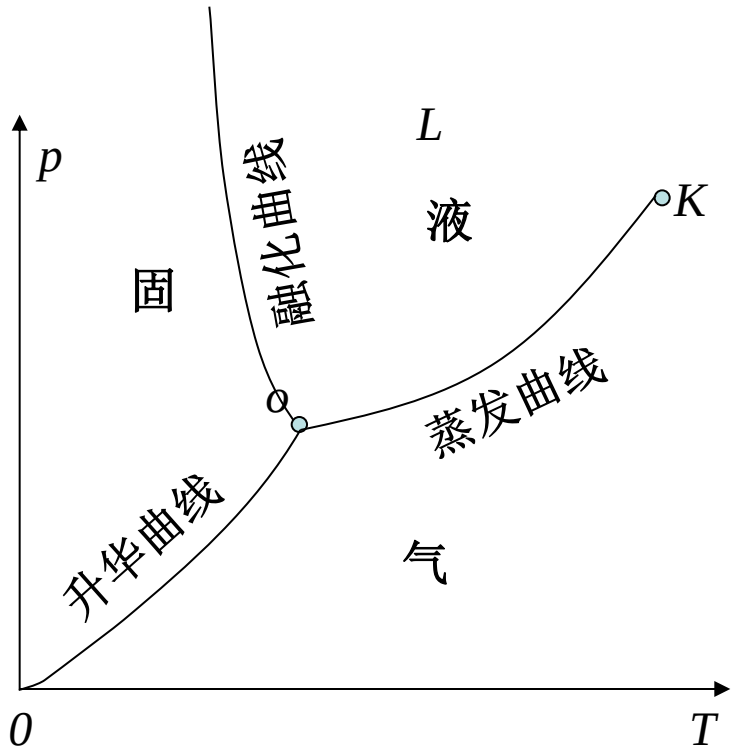
$$\frac{p}{p_0} = \exp[-\frac{l_m}{R}(\frac{1}{T} - \frac{1}{T_0})] \quad \sim 0.5$$

地面 $p_0 = 1\text{atm}$ $\quad l_m = 4.07 \times 10^4 \text{J/mol}$

$T_0 = 373\text{K}$ $\quad T = 353\text{K}$

冲击波过后压强陡降,
潮湿的天气,有水汽凝结

冰的融解反常现象



$$\frac{dp}{dT} < 0$$

滚雪球

冰川

滑雪、滑冰？

冰为什么滑?

$$\frac{dp}{dT} = \frac{l_{v,m}}{R} \cdot \frac{p}{T^2}$$

$$\frac{dT}{dp} = -0.074\text{K/MPa}$$

标准状态下

100公斤人，冰刀面积 1mm×30cm $\Delta T \approx -0.23^{\circ}\text{C}$

1859年 Michael Faraday

即时是在零点以下温度， 冰面总有一薄层水

薄水层随温度下降逐渐消失

谢谢！